PROJECT: E-Commerce Shopping Cart System (Java - Console Based)

1. System Overview (Simple Understanding)

You are building a system like:
 Amazon / Flipkart (basic version)

Where user can:

2. View products
3. Add to cart
4. Apply coupon
5. Checkout
6. System updates stock

2. OOP DESIGN

class Product {

int id;

String name;

double price;

int stock;

}

1. Product

```
class Product {
    int id;
    String name;
    double price;
    int stock;
}
```

---

2. CartItem

```
class CartItem {
    Product product;
    int quantity;
}
```

---

3. Cart

```
class Cart {
    ArrayList<CartItem> items;

    void addItem(Product p, int qty) {}
    void removeItem(int productId) {}
    void updateQuantity(int productId, int qty) {}
    double calculateSubtotal() {}
}
```

---

4. Coupon

```
class Coupon {
    String code;
    String type; // "PERCENT" or "FLAT"
    double value;
}
```

---

5. Order

```
class Order {
    ArrayList<CartItem> items;
    double totalAmount;
}
```

---

6. Main Class

Handles:

- Menu
- User input
- Flow control

---

3. FLOW OF PROGRAM

```
Start Program
   ↓
Load Products (from CSV)
   ↓
Show Menu
```

```
   ↓
1. View Products
2. Add to Cart
3. View Cart
4. Apply Coupon
5. Checkout
   ↓
Process Order
   ↓
Update Stock
   ↓
Show Bill
```

---

4. DATA STORAGE (CSV)

products.csv

```
1,Laptop,50000,10
2,Mouse,500,50
3,Keyboard,2000,20
```

coupons.csv

```
SAVE20,PERCENT,20
FLAT500,FLAT,500
```

---

5. CORE LOGIC (IMPORTANT)

Add to Cart

- Check stock
- If already in cart → increase qty
- Prevent negative or zero qty

Discount Logic

- Percentage → `(subtotal * value / 100)`
- Flat → subtract directly

Tax Calculation

```
Tax = 10% of (discounted price)
Final = discounted + tax
```

Checkout

- Check stock again
- Deduct stock
- Generate bill

□ 6. BASIC STARTER CODE

Here is a **working base version (simplified)** □

```
import java.util.*;

class Product {
    int id;
    String name;
    double price;
    int stock;

    Product(int id, String name, double price, int stock) {
        this.id = id;
        this.name = name;
        this.price = price;
        this.stock = stock;
    }
}

class CartItem {
    Product product;
    int quantity;

    CartItem(Product product, int quantity) {
        this.product = product;
        this.quantity = quantity;
    }
}

class Cart {
    ArrayList<CartItem> items = new ArrayList<>();

    void addItem(Product p, int qty) {
        if (qty <= 0) {
            System.out.println("Quantity must be positive");
```

```
            return;
        }

        for (CartItem item : items) {
            if (item.product.id == p.id) {
                item.quantity += qty;
                return;
            }
        }
        items.add(new CartItem(p, qty));
    }

    void viewCart() {
        for (CartItem item : items) {
            System.out.println(item.product.name + " x " + item.quantity);
        }
    }

    double getSubtotal() {
        double total = 0;
        for (CartItem item : items) {
            total += item.product.price * item.quantity;
        }
        return total;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        ArrayList<Product> products = new ArrayList<>();
        products.add(new Product(1, "Laptop", 50000, 10));
        products.add(new Product(2, "Mouse", 500, 50));

        Cart cart = new Cart();

        while (true) {
            System.out.println("\n1. View Products\n2. Add to Cart\n3. View
Cart\n4. Checkout");
            int choice = sc.nextInt();

            if (choice == 1) {
                for (Product p : products) {
                    System.out.println(p.id + " " + p.name + " ₹" + p.price +
" Stock:" + p.stock);
                }
            }

            else if (choice == 2) {
                System.out.print("Enter Product ID: ");
                int id = sc.nextInt();
                System.out.print("Enter Quantity: ");
                int qty = sc.nextInt();

                for (Product p : products) {
                    if (p.id == id) {
```

```
                        if (qty > p.stock) {
                            System.out.println("Insufficient stock!");
                        } else {
                            cart.addItem(p, qty);
                        }
                    }
                }
            }

            else if (choice == 3) {
                cart.viewCart();
                System.out.println("Subtotal: ₹" + cart.getSubtotal());
            }

            else if (choice == 4) {
                double subtotal = cart.getSubtotal();
                double tax = subtotal * 0.10;
                double total = subtotal + tax;

                System.out.println("Final Amount: ₹" + total);
                break;
            }
        }
    }
}
```